

2025-11-06 - Valentini (11)

Previous lecture: [2025-10-31 - Valentini \(10\)](#)

Next lecture: [2025-11-07 - Valentini \(12\)](#)

Recap

$$\omega_R = \frac{k c_s}{\sqrt{1 + k^2 \lambda_{De}^2}} \left[1 + 3 \frac{T_{i0}}{T_{e0}} \right]^{1/2}$$

with

$$\begin{aligned} \left| \frac{\omega_I}{\omega_R} \right| &\ll 1, \quad \frac{T_{i0}}{T_{e0}} \ll 1 \\ D_R(k, \omega_R) &= 1 - \frac{\omega_{pi}^2}{\omega_R^2} \left[1 + \frac{3k^2 \lambda_{Di}^2 \omega_{pi}^2}{\omega_R^2} \right] + \frac{1}{k^2 \lambda_{De}^2} \\ \omega_I &= - \frac{D_I(k, \omega_R)}{\partial D_R / \partial \omega_R} \\ D_I(k, \omega_R) &= -\pi \sum_{\alpha} \frac{\omega_{p\alpha}^2}{k^2 n_{\alpha 0}} \frac{\partial F_{\alpha 0}(u)}{\partial u} \Big|_{\omega_R/k} = \\ &= -\pi \frac{\omega_{pi}^2}{k^2 n_0} \frac{\partial F_{i0}(u)}{\partial u} \Big|_{\omega_R/k} - \pi \frac{\omega_{pe}^2}{k^2 n_0} \frac{\partial F_{e0}(u)}{\partial u} \Big|_{\omega_R/k} \end{aligned}$$

In the low frequency regime we also have

$$v_{th i} \ll \frac{\omega_R}{k} \ll v_{th e}$$

We use the approximation

$$D_R(k, \omega_R) \approx 1 - \frac{\omega_{pi}^2}{\omega_R^2} + \frac{1}{k^2 \lambda_{De}^2} \implies \frac{\partial D_R}{\partial \omega_R} \approx \frac{2\omega_{pi}^2}{\omega_R^3}$$

Now we want to find ω_I

$$\begin{aligned} \omega_I &= \frac{\pi}{k^2 n_0} \frac{\omega_R^3}{2\omega_{pi}^2} \left[\omega_{pi}^2 \frac{\partial F_{i0}(u)}{\partial u} \Big|_{\omega_R/k} + \omega_{pe}^2 \frac{\partial F_{e0}(u)}{\partial u} \Big|_{\omega_R/k} \right] = \\ &= \frac{\pi}{k^2 n_0} \frac{\omega_R^3}{2} \left[\frac{\partial F_{i0}(u)}{\partial u} \Big|_{\omega_R/k} + \frac{\omega_{pe}^2}{\omega_{pi}^2} \frac{\partial F_{e0}(u)}{\partial u} \Big|_{\omega_R/k} \right] \end{aligned}$$

Now we use the maxwellian distribution and its derivative

$$F_{\alpha 0}(u) = \frac{n_0 m_\alpha^{1/2}}{(2\pi k_B T_{\alpha 0})^{1/2}} \exp\left(-\frac{m_\alpha}{2k_B T_{\alpha 0}} u^2\right)$$

$$\frac{\partial F_{\alpha 0}(u)}{\partial u} = \frac{n_0 m_\alpha^{1/2}}{(2\pi k_B T_{\alpha 0})^{1/2}} \left(-\frac{m_\alpha u}{k_B T_{\alpha 0}}\right) \exp\left(-\frac{m_\alpha}{2k_B T_{\alpha 0}} u^2\right)$$

$$\left.\frac{\partial F_{\alpha 0}(u)}{\partial u}\right|_{\omega_R/k} = -\frac{n_0 m_\alpha^{3/2}}{(2\pi)^{1/2} (k_B T_{\alpha 0})^{3/2}} \frac{\omega_R}{k} \exp\left(-\frac{m_\alpha}{2k_B T_{\alpha 0}} \frac{\omega_R^2}{k^2}\right)$$

and

$$\begin{aligned} \omega_I &= \frac{\pi}{k^2 n_0} \frac{\omega_R^3}{2} \left[-\frac{n_0 m_i^{3/2}}{(2\pi)^{1/2} (k_B T_{i0})^{3/2}} \frac{\omega_R}{k} \exp\left(-\frac{m_i}{2k_B T_{i0}} \frac{\omega_R^2}{k^2}\right) + \right. \\ &\quad \left. - \frac{\omega_{pe}^2}{\omega_{pi}^2} \frac{n_0 m_e^{3/2}}{(2\pi)^{1/2} (k_B T_{e0})^{3/2}} \frac{\omega_R}{k} \exp\left(-\frac{m_e}{2k_B T_{e0}} \frac{\omega_R^2}{k^2}\right) \right] = \\ &= -\sqrt{\frac{\pi}{8}} \frac{\omega_R^4 m_i^{3/2}}{k^3 (k_B T_{e0})^{3/2}} \left[\left(\frac{T_{e0}}{T_{i0}}\right)^{3/2} \exp\left(-\frac{m_i}{2k_B T_{i0}} \frac{\omega_R^2}{k^2}\right) + \right. \\ &\quad \left. + \frac{m_i}{m_e} \left(\frac{m_e}{m_i}\right)^{3/2} \exp\left(-\frac{m_e}{2k_B T_{e0}} \frac{\omega_R^2}{k^2}\right) \right] = \\ &= -\sqrt{\frac{\pi}{8}} \frac{\omega_R^4 m_i^{3/2}}{k^3 (k_B T_{e0})^{3/2}} \left[\left(\frac{T_{e0}}{T_{i0}}\right)^{3/2} \exp\left(-\frac{m_i}{2k_B T_{i0}} \frac{\omega_R^2}{k^2}\right) + \right. \\ &\quad \left. + \left(\frac{m_e}{m_i}\right)^{1/2} \exp\left(-\frac{m_e}{2k_B T_{e0}} \frac{\omega_R^2}{k^2}\right) \right] \end{aligned}$$

Now, using

$$\frac{T_{i0}}{T_{e0}} \ll 1 \quad \text{and} \quad \frac{m_e}{m_i} \ll 1$$

we approximate

$$\exp\left(-\frac{m_e}{2k_B T_{e0}} \frac{\omega_R^2}{k^2}\right) \approx 1$$

and

$$\begin{aligned} \omega_I &= -\sqrt{\frac{\pi}{8}} \frac{\omega_R^4}{k^3 c_s^3} \left[\left(\frac{T_{e0}}{T_{i0}}\right)^{3/2} \exp\left(-\frac{m_i}{2k_B T_{i0}} \frac{\omega_R^2}{k^2}\right) + \sqrt{\frac{m_e}{m_i}} \right] = \\ &= -\sqrt{\frac{\pi}{8}} \frac{\omega_R}{\cancel{k^3 c_s^3}} \frac{\cancel{k^3 c_s^3}}{(1 + k^2 \lambda_{De}^2)^{3/2}} \left[\left(\frac{T_{e0}}{T_{i0}}\right)^{3/2} \exp\left(-\frac{m_i}{2k_B T_{i0}} \frac{\omega_R^2}{k^2} \cdot \frac{T_{e0}}{T_{e0}}\right) + \sqrt{\frac{m_e}{m_i}} \right] = \\ &= -\sqrt{\frac{\pi}{8}} \frac{\omega_R}{(1 + k^2 \lambda_{De}^2)^{3/2}} \left[\left(\frac{T_{e0}}{T_{i0}}\right)^{3/2} \exp\left(-\frac{T_{e0}}{2T_{i0}} \frac{1}{1 + k^2 \lambda_{De}^2}\right) + \sqrt{\frac{m_e}{m_i}} \right] \end{aligned}$$

where in the second passage we used

$$\omega_R = \frac{k^3 c_s^3}{(1 + k^2 \lambda_{De}^2)^{3/2}} \left[1 + 3 \frac{T_{i0}}{T_{e0}} \right]^{1/2} \approx \frac{k^3 c_s^3}{(1 + k^2 \lambda_{De}^2)^{3/2}}$$

In the last expression of ω_I we have the factor

$$\sqrt{\frac{m_e}{m_i}} \sim \frac{1}{40}$$

which is negligible.

Furthermore we can affirm that

$$\frac{T_{i0}}{T_{e0}} \ll 1 \quad \Rightarrow \quad \left| \frac{\omega_I}{\omega_R} \right| \ll 1$$

which in conclusion represent the same condition.

Bump-on-tail instability

$$D(k, \omega) = 1 - \sum_{\alpha} \frac{\omega_{p\alpha}^2}{k^2 n_{\alpha 0}} \int_L \frac{\partial F_{\alpha 0}(u)/\partial u}{u - \omega/k} du$$

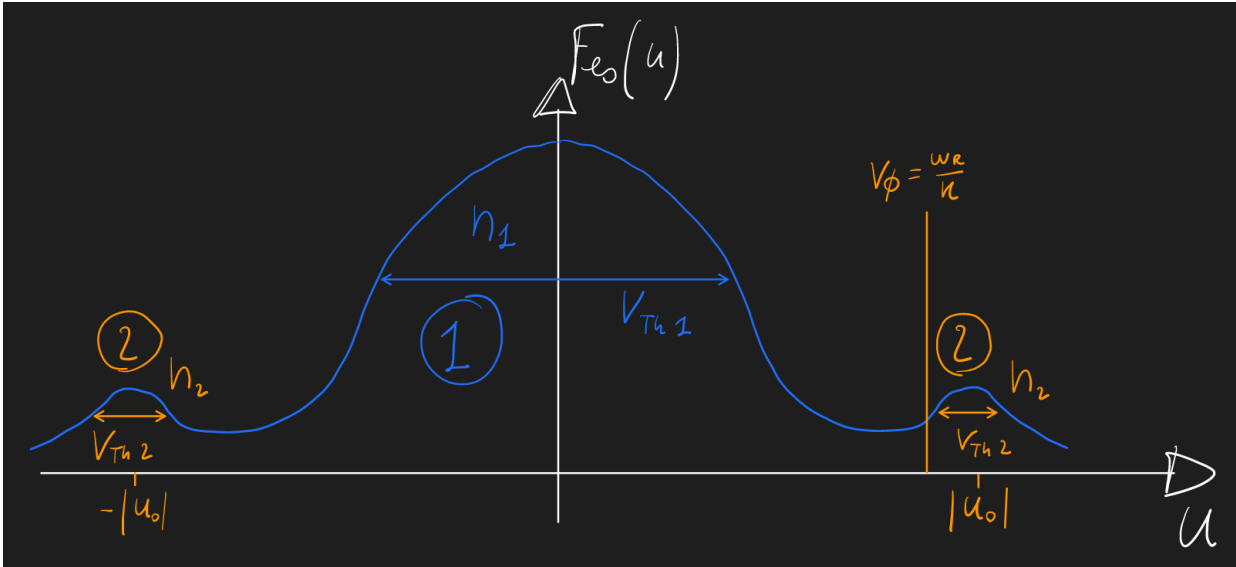
$$D_R(k, \omega_R) = 1 - \sum_{\alpha} \frac{\omega_{p\alpha}^2}{k^2 n_{\alpha 0}} P \int_{-\infty}^{\infty} \frac{\partial F_{\alpha 0}(u)/\partial u}{u - \omega_R/k} du$$

$$D_I(k, \omega_R) = -\pi \sum_{\alpha} \frac{\omega_{p\alpha}^2}{k^2 n_{\alpha 0}} \left. \frac{\partial F_{\alpha 0}(u)}{\partial u} \right|_{\omega_R/k}$$

High frequency regime

Propagation speed is too high for ions to follow the wave, so only electrons give a contribute.

PIC BumpOnTail



These symmetric bumps avoid the generation a current $\bar{J}$, hence also the magnetic field will be null (this is required by the electrostatic approximation in the derivation of Vlasov-Poisson equations).

$$|u_0| \gg v_{th1} = \sqrt{\frac{k_B T_1}{m_e}}, \quad \frac{\omega_R}{k} \gg v_{th1} \gg v_{th2}, \quad T_1 \gg T_2$$

"gentle bump"

We also assume

$$n_1 + 2n_2 \approx n_1, \quad \frac{n_2}{n_1} \ll 1$$

and

$$k^2 \lambda_{De}^2 \ll 1, \quad \omega_I \propto \left. \frac{\partial F_{e0}(u)}{\partial u} \right|_{\omega_R/k}$$

We use the following distribution function

$$\begin{aligned} F_{e0}(u) = & \frac{m_e^{1/2} n_1}{(2\pi k_B T_1)^{1/2}} \exp\left(-\frac{m_e}{2k_B T_1} u^2\right) + \\ & + \frac{m_e^{1/2} n_2}{(2\pi k_B T_2)^{1/2}} \exp\left(-\frac{m_e}{2k_B T_2} (u - |u_0|)^2\right) + \\ & + \frac{m_e^{1/2} n_2}{(2\pi k_B T_2)^{1/2}} \exp\left(-\frac{m_e}{2k_B T_2} (u + |u_0|)^2\right) \end{aligned}$$

In the integral of $D_R(k, \omega_R) = 0$ we can neglect the derivative because it is local, the high value of the derivatives is due to the bumps, but it is negligible with respect to the rest of the integral.

$$D_R(k, \omega_R) = 0 \quad \implies \quad \omega_R = \omega_{p1} \sqrt{1 + 3k^2 \lambda_{D1}^2}$$

and

$$D_R(k, \omega_R) = 1 - \frac{\omega_{p1}^2}{\omega_R^2} - \frac{3k^2 \lambda_{D1}^2 \omega_{p1}^2}{\omega_R^2} = 0$$

when we find the derivative of such quantity, we will neglect the last term ($k^2 \lambda_{D1}^2 \ll 1$) and

$$\frac{\partial D_R}{\partial \omega_R} \approx \frac{2\omega_{p1}^2}{\omega_R^3}$$

For the imaginary part, we cannot neglect the local derivative at the bumps.

First we evaluate

$$\begin{aligned} \frac{\partial F_{e0}(u)}{\partial u} = & \frac{m_e^{3/2} n_1}{(2\pi)^{1/2} (k_B T_1)^{3/2}} u \exp\left(-\frac{m_e}{2k_B T_1} u^2\right) + \\ & - \frac{m_e^{3/2} n_2}{(2\pi)^{1/2} (k_B T_2)^{3/2}} \left\{ (u - |u_0|) \exp\left(-\frac{m_e}{2k_B T_2} (u - |u_0|)^2\right) + \right. \\ & \left. + (u + |u_0|) \exp\left(-\frac{m_e}{2k_B T_2} (u + |u_0|)^2\right) \right\} \end{aligned}$$

Now we evaluate this quantity in $u = \omega_R/k$.

We notice that the third exponential is similar to $\exp(-M)$ with M a really big number, so this term tends to zero and we neglect it.

We have

$$\begin{aligned} \left. \frac{\partial F_{e0}(u)}{\partial u} \right|_{\omega_R/k} &= \frac{m_e^{3/2} n_1}{(2\pi)^{1/2} (k_B T_1)^{3/2}} \frac{\omega_R}{k} \exp \left[-\frac{m_e}{2k_B T_1} \left(\frac{\omega_R}{k} \right)^2 \right] + \\ &\quad - \frac{m_e^{3/2} n_2}{(2\pi)^{1/2} (k_B T_2)^{3/2}} \left(\frac{\omega_R}{k} - |u_0| \right) \exp \left[-\frac{m_e}{2k_B T_2} \left(\frac{\omega_R}{k} - |u_0| \right)^2 \right] \end{aligned}$$

So

$$\begin{aligned} \omega_I &= -\frac{D_I}{\partial D_R / \partial \omega_R} = \\ &= (-1)(-1) \frac{\omega_R^3}{2\omega_{p1}^2} \pi \left\{ -\frac{\omega_{p1}^2}{k^2} \frac{m_e^{3/2} \cancel{\omega_R}}{k(2\pi)^{1/2} (k_B T_1)^{3/2}} \exp \left[-\frac{m_e}{2k_B T_1} \left(\frac{\omega_R}{k} \right)^2 \right] + \right. \\ &\quad \left. - \frac{m_e^{3/2} \cancel{\omega_R} \omega_{p2}^2}{k^2 \cancel{\omega_R} (2\pi)^{1/2} (k_B T_2)^{3/2}} \left(\frac{\omega_R}{k} - |u_0| \right) \exp \left[-\frac{m_e}{2k_B T_1} \left(\frac{\omega_R}{k} - |u_0| \right)^2 \right] \right\} = \\ &= \omega_R^3 \sqrt{\frac{\pi}{8}} \frac{\cancel{\omega_{p1}^2}}{\omega_{p1}^2} \frac{m_e^{3/2}}{k^2 (k_B T_1)^{3/2}} \left\{ -\frac{\omega_R}{k} \exp \left[-\frac{m_e}{2k_B T_1} \left(\frac{\omega_R}{k} \right)^2 \right] + \right. \\ &\quad \left. + \frac{n_2}{n_1} \left(\frac{T_1}{T_2} \right)^{3/2} \left(|u_0| - \frac{\omega_R}{k} \right) \exp \left[-\frac{m_e}{2k_B T_2} \left(|u_0| - \frac{\omega_R}{k} \right)^2 \right] \right\} \end{aligned}$$

Where we used the fact that

$$\frac{\omega_R}{k} < |u_0|$$

Now we collect ω_R/k and

$$\begin{aligned} \omega_I &= \sqrt{\frac{\pi}{8}} \frac{\omega_R^4 m_e^{3/2}}{k^3 (k_B T_1)^{3/2}} \left\{ -\exp \left[-\frac{m_e}{2k_B T_1} \left(\frac{\omega_R}{k} \right)^2 \right] + \right. \\ &\quad \left. + \frac{n_2}{n_1} \left(\frac{T_1}{T_2} \right)^{3/2} \left(\frac{|u_0|k}{\omega_R} - 1 \right) \exp \left[-\frac{m_e}{2k_B T_2} \frac{\omega_R^2}{k^2} \left(\frac{|u_0|k}{\omega_R} - 1 \right)^2 \right] \right\} = \\ &= \sqrt{\frac{\pi}{8}} \frac{\omega_R \omega_{p1}^3}{k^3 v_{th1}^3} \left\{ -\exp \left[-\frac{m_e}{2k_B T_1} \frac{\omega_{p1}^2 (1 + 3k^2 \lambda_{D1}^2)}{2} \right] + \right. \\ &\quad \left. + \frac{n_2}{n_1} \left(\frac{T_1}{T_2} \right)^{3/2} \left(\frac{|u_0|k}{\omega_R} - 1 \right) \exp \left[-\frac{m_e}{2k_B T_2} \frac{T_1}{T_1} \frac{\omega_{p1}^2}{k^2} \left(\frac{|u_0|k}{\omega_R} - 1 \right)^2 \right] \right\} = \\ &= \sqrt{\frac{\pi}{8}} \frac{\omega_R}{k^3 \lambda_{D1}^3} \left\{ -\exp \left[-\frac{1}{2k^2 \lambda_{D1}^2} - \frac{3}{2} \right] + \right. \\ &\quad \left. + \frac{n_2}{n_1} \left(\frac{T_1}{T_2} \right)^{3/2} \left(\frac{|u_0|k}{\omega_R} - 1 \right) \exp \left[-\frac{1}{2k^2 \lambda_{D1}^2} \frac{T_1}{T_2} \left(\frac{|u_0|k}{\omega_R} - 1 \right)^2 \right] \right\} \end{aligned}$$

We isolate

$$\omega_{I,\text{beam}} = \underbrace{\sqrt{\frac{\pi}{8}} \frac{\omega_R}{k^3 \lambda_{D1}^3} \frac{n_2}{n_1} \left(\frac{T_1}{T_2}\right)^{3/2}}_{\beta} \left(\frac{|u_0|k}{\omega_R} - 1 \right) \exp \left[-\frac{1}{2k^2 \lambda_{D1}^2} \frac{T_1}{T_2} \left(\frac{|u_0|k}{\omega_R} - 1 \right)^2 \right]$$

We use the change of variable

$$y = \frac{|u_0|k}{\omega_R}$$

and we want to find the maximum of this quantity

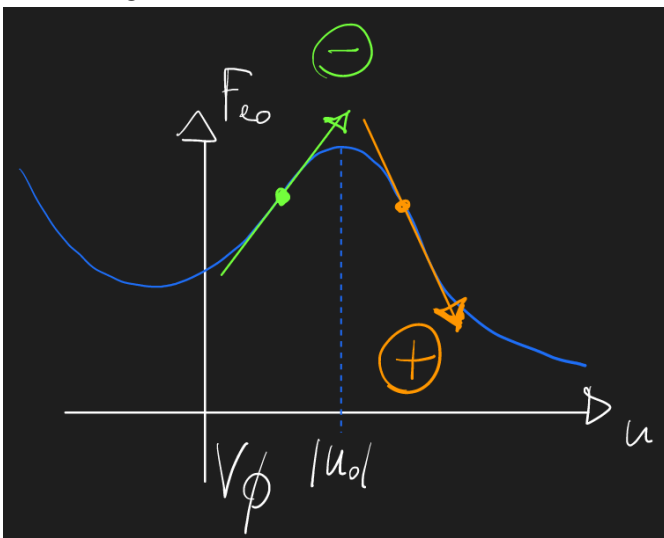
$$\begin{aligned} \frac{\partial \omega_{I,\text{beam}}}{\partial y} &= \frac{\partial}{\partial y} \left\{ \beta(y-1) \exp \left[-\frac{1}{2k^2 \lambda_{D1}^2} \frac{T_1}{T_2} (y-1)^2 \right] \right\} = \\ &= \beta \exp \left[-\frac{1}{2k^2 \lambda_{D1}^2} \frac{T_1}{T_2} (y-1)^2 \right] + \\ &\quad + \beta(y-1) \left(-\frac{1}{k^2 \lambda_{D1}^2} \frac{T_1}{T_2} (y-1) \right) \exp \left[-\frac{1}{2k^2 \lambda_{D1}^2} \frac{T_1}{T_2} (y-1)^2 \right] = \\ &= \beta \left(1 - \frac{(y-1)^2}{k^2 \lambda_{D1}^2} \frac{T_1}{T_2} \right) \exp \left[-\frac{1}{2k^2 \lambda_{D1}^2} \frac{T_1}{T_2} (y-1)^2 \right] = 0 \end{aligned}$$

We know that $\exp(x) \neq 0, \forall x \in \mathbb{R}$ so

$$\begin{aligned} 1 - \frac{(y-1)^2}{k^2 \lambda_{D1}^2} \frac{T_1}{T_2} &= 0 & \implies (y-1)^2 &= \frac{T_2}{T_1} k^2 \lambda_{D1}^2 \\ \implies y-1 &= \pm \sqrt{\frac{T_2}{T_1}} k \lambda_{D1} & \implies y &= 1 \pm \sqrt{\frac{T_2}{T_1}} k \lambda_{D1} \\ \implies \frac{|u_0|k}{\omega_R} &= 1 \pm \sqrt{\frac{T_2}{T_1}} k \lambda_{D1} & \implies |u_0| &= \frac{\omega_R}{k} \left(1 \pm \sqrt{\frac{T_2}{T_1}} k \lambda_{D1} \right) \end{aligned}$$

Between these two solution of $|u_0|$ we choose the one which satisfies the instability condition ($\omega_I > 0$) and which could lead to instabilities.

PIC der .g. zero



We substitute this definition into ω_I and

$$\begin{aligned}\omega_I &= \sqrt{\frac{\pi}{8}} \frac{\omega_R}{k^3 \lambda_{D1}^3} \left\{ -\exp \left[-\frac{1}{2k^2 \lambda_{D1}^2} - \frac{3}{2} \right] + \right. \\ &\quad \left. + \frac{n_2}{n_1} \left(\frac{T_1}{T_2} \right)^{3/2} \left(1 - \sqrt{\frac{T_2}{T_1}} k \lambda_{D1} - 1 \right) \exp \left[-\frac{1}{2k^2 \lambda_{D1}^2} \frac{T_1}{T_2} \frac{T_2}{T_1} k^2 \lambda_{D1}^2 \right] \right\} = \\ &= \sqrt{\frac{\pi}{8}} \frac{\omega_R}{k^3 \lambda_{D1}^3} \left\{ \frac{n_2}{n_1} \frac{T_1}{T_2} k \lambda_{D1} e^{-1/2} - \exp \left[-\frac{1}{2k^2 \lambda_{D1}^2} - \frac{3}{2} \right] \right\} > 0\end{aligned}$$

Which is the instability condition.

To have an instability it is sufficient to have the maximum point as an unstable point.

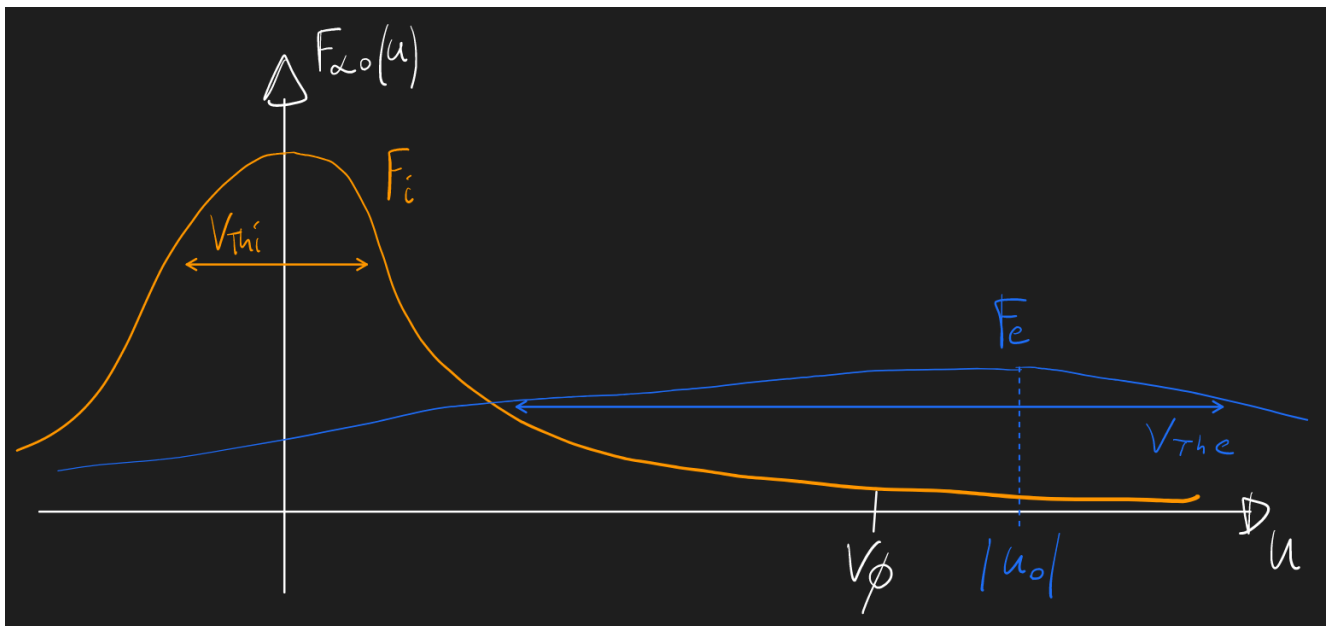
It's possible to repeat this for the Ion Acoustic Waves, with the following hypothesis

$$\left| \frac{\omega_I}{\omega_R} \right| \ll 1, \quad \frac{\omega_R}{k} \gg v_{thi}, \quad \frac{T_e}{T_i} \gg 1$$

and

$$\left| \frac{\omega_R}{k} - u_0 \right| \ll v_{the}, \quad (u_0 > 0)$$

PIC - ion acoustic wave distribution function



Previous lecture: [2025-10-31 - Valentini \(10\)](#).

Next lecture: [2025-11-07 - Valentini \(12\)](#).